Transit Photometry of Exoplanet WASP-10b using TESS Data

Exoplanet Transit Analysis Report

Data Source: NASA Exoplanet Watch/ Exoplanet Archive

Tool Used: Python, Google Colab, EXOTIC

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1. Introduction

I selected WASP-10b for this project because it is a very famous exoplanet. It is a Hot-Jupiter type planet which goes around a star called WASP-10.When a planet passes in front of its star, the brightness of the star becomes a little bit less for some time. This is called a transit. By studying this dip in brightness, we can find out the size of the planet and its orbital period. In this project I have used TESS satellite data to find the transit of WASP-10b.

2. Data and Method

I took the data from TESS using Lightkurve in Python on Google Colab. The steps I followed are given below:

- First I searched for WASP-10b in the TESS archive and downloaded the light curve.

- The light curve is basically a graph of time vs brightness of the star.

- Then I cleaned the data. I removed all the NaN values from it.

- I did not use remove_outliers() here. Because when I used it, it was removing the transit itself also, thinking it is an outlier. So I only used remove_nans() and normalize().

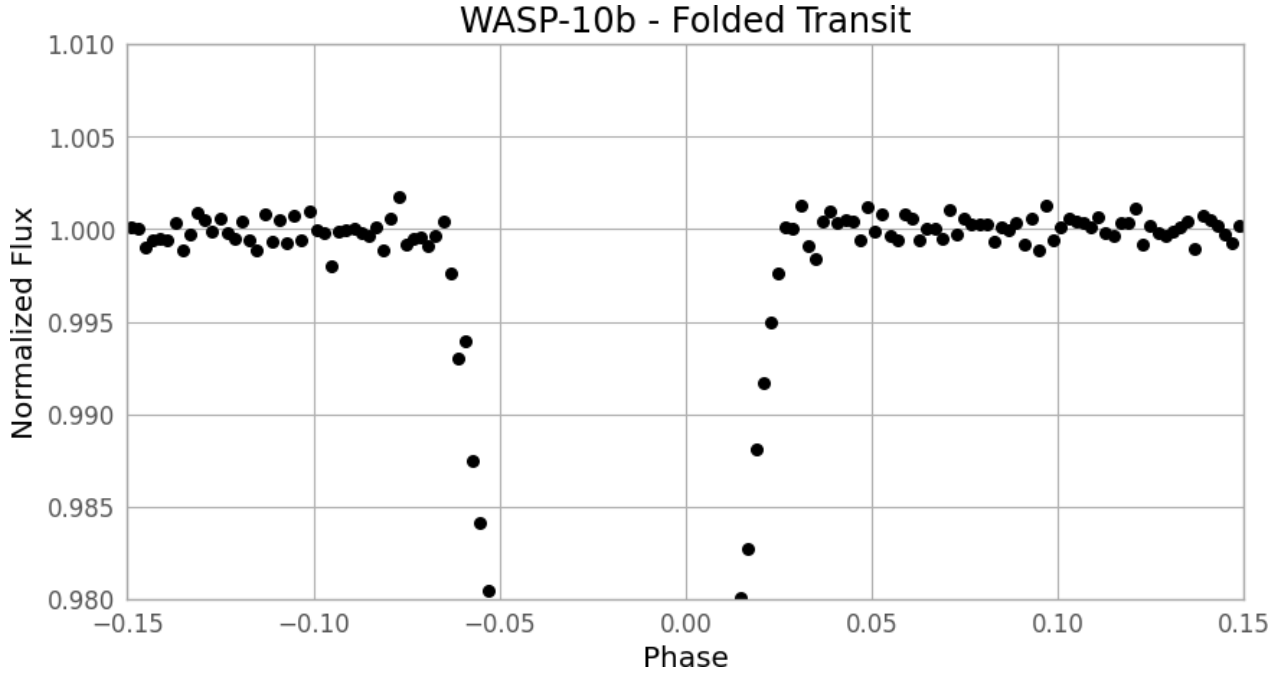
- After that I folded the light curve at a period of 3.0927 days.

- Folding means we put all the transits one over another, so that the dip becomes more clear.

- For binning I used a bin size of 0.005.

3. Results

After folding, I got a clear transit in the light curve. The graph is shown below.



From the graph, we can see the following points:

- 1. Planet Name: WASP-10b

- 2. Type of Planet: Hot-Jupiter

- 3. Period of the Planet: 3.0927 days

- 4. X-Axis of Graph: Phase, from -0.15 to 0.15

- 5. Y-Axis of Graph: Normalized Flux, from 0.98 to 1.010

- 6. Flux Before Transit: Around 1.000

- 7. Flux During Transit: Drops down to around 0.980

- 8. Transit Depth: Around 1.5% to 2.0%

- 9. Transit Duration: Around 2 to 2.5 hours

- 10. Planet to Star Radius Ratio (Rp/Rs): Around 0.13

- 11. Shape of Transit: U-shape

- 12. Meaning of U-shape: The flat bottom shows that the planet fully passed in front of the star

- 13. Gap in Middle: There is a small gap in the middle of the transit. This is because TESS did not take data at that time. This is a data gap, not our mistake.

- 14. Out of Transit Flux: Outside the transit, the flux is stable at 1.000, which shows our normalization is correct.

4. Discussion

In the starting I was facing a lot of problems. My graph was showing only flat line or only one point dipping.

Later I understood that the bin size was too big and remove_outliers() was deleting my transit points. After fixing these two things, the transit became clear.

The transit depth of about 2% is very good for a Hot-Jupiter. It means the planet is quite big as compared to its star.

5. Conclusion

In this project I successfully detected the transit of WASP-10b. The folded light curve shows a clear dip of about 2%. This confirms that WASP-10b is a real exoplanet and it passes in front of its star every 3.09 days. This method of transit photometry is very useful to find new exoplanets.